Take-Home Part II

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Question 1

A major difficulty in what we did previously was determining a good covariance matrix to use in the construction of confidence intervals based on asymptotic normality, regardless of whether that asymptotic normality was based on maximum likelihood or maximum composite likelihood estimation. One possible alternative to the direct use of asymptotic results is to compute intervals directly using parametric bootstrap methods.

(a)

> Determine 95% confidence intervals for the five parameters of the model through the application of percentile bootstrap intervals for maximum likelihood estimation.

The 95% percentile bootstrap confidence intervals for all five parameters are reported in Table 1; the construction is described below. This approach avoids reliance on the observed information matrix, which was numerically unstable in Part I.

Methodology. We construct 95% percentile bootstrap confidence intervals for each of the five model parameters $(\alpha, \beta, \gamma, \theta, \sigma^2)$ following the development in Section 10.3. Following expression (10.15), the $(1-\alpha)100\%$ percentile bootstrap interval for a scalar parameter is given by the empirical $\alpha/2$ and $1-\alpha/2$ quantiles of the M bootstrap estimates — no discrepancy function need be formed or inverted, and no symmetrizing transformation need be determined explicitly.

Simulation from the fitted model. We take $M = 999$ so that $(M+1)\cdot 0.025 = 25$ is an integer, satisfying the integrality condition noted after expression (10.10). For each bootstrap replicate $m = 1, \ldots, M$, we simulate $\mathbf{y}_m^* = (y_1^*, \ldots, y_n^*)^T$ from the fitted model $F(\mathbf{y} \mid \hat{\theta}_n)$ — the model estimated at the MLE from the observed data, consistent with the general simulation framework described in Section 10.1. Because the model includes AR(1) errors, simulation requires drawing $\varepsilon_1^* \sim N\Big(0,\, \hat{\sigma}^2/(1-\hat{\theta}^2)\Big)$ from the stationary distribution and then propagating $\varepsilon_i^* = \hat{\theta}\,\varepsilon_{i-1}^* + \delta_i$ with $\delta_i \overset{iid}{\sim} N(0, \hat{\sigma}^2)$ for $i = 2, \ldots, n$. The simulated response is then $y_i^* = \hat{\mu}_i + \varepsilon_i^*$ where $\hat{\mu}_i = \hat{\alpha}\, x_i^{\hat{\beta} x_i^{-\hat{\gamma}}}$.

Re-estimation. Each bootstrap dataset $\mathbf{y}_m^*$ is re-fit by maximizing the same log-likelihood as in Q3, using the original MLE $\hat{\theta}_n$ as the starting point for BFGS optimization. This yields bootstrap estimates $\theta_{n,m}^*$ for $m = 1, \ldots, M$ in the notation of Section 10.1.

Remark (Convergence filtering). The parametric bootstrap approximates the sampling distribution of the estimator as a *procedure*, not merely as a formal functional of the data. The estimator is defined implicitly through a numerical optimization routine, which may fail to converge for certain bootstrap datasets; for such replicates the estimator fails to exist numerically, and they are therefore discarded. We do not resample to replace them, as doing so would alter the bootstrap data-generating mechanism by preferentially retaining datasets for which the likelihood surface is sufficiently regular, potentially excluding replicates corresponding to extreme parameter estimates and biasing the empirical distribution toward regions of the sample space where estimation is numerically stable.

Interval construction. Of the $M = 999$ replicates, $M_{\text{eff}} = 988$ converged successfully and were retained (using `set.seed(42)`). Inference is therefore based on the empirical distribution of $\hat{\theta}^*_{n,m}$ conditional on successful convergence of the re-estimation procedure applied to $\mathbf{y}^*_m$. Following expression (10.15), the 95% percentile interval for each parameter is the empirical 2.5% and 97.5% quantiles of the $M_{\text{eff}} = 988$ bootstrap estimates, computed via interpolation rather than exact order-statistic indexing since $(M_{\text{eff}} + 1) \cdot 0.025 = 24.725$ is no longer an integer. This conditioning may modestly understate uncertainty, but with only 11 failures out of 999 the effect is expected to be negligible. No discrepancy function $h(\theta, \hat{\theta}_n)$ needs to be formed or inverted — unlike the basic interval (10.10) or studentized interval (10.12) — because the percentile method works directly on the empirical distribution of $\theta^*_{n,m}$.

The asymptotic-normal intervals from Q3 are included in Table 1 for comparison. The concerns of Part I regarding whether the relevant asymptotics hold are not eliminated by this approach, but the bootstrap avoids the need to invert the observed information matrix directly.

Table 1: 95% Percentile Bootstrap and Asymptotic-Normal CIs

Parameter	MLE	Boot 2.5%	Boot 97.5%	AN 2.5%	AN 97.5%
α	0.7395	0.0675	2.5107	-0.4834	1.9625
β	1.0283	0.3085	2.5010	0.0275	2.0291
γ	0.1574	0.0552	0.2099	0.0929	0.2219
θ	0.5598	-0.1289	0.6488	0.2203	0.8993
σ^2	0.6250	0.2633	0.8790	0.2785	0.9714

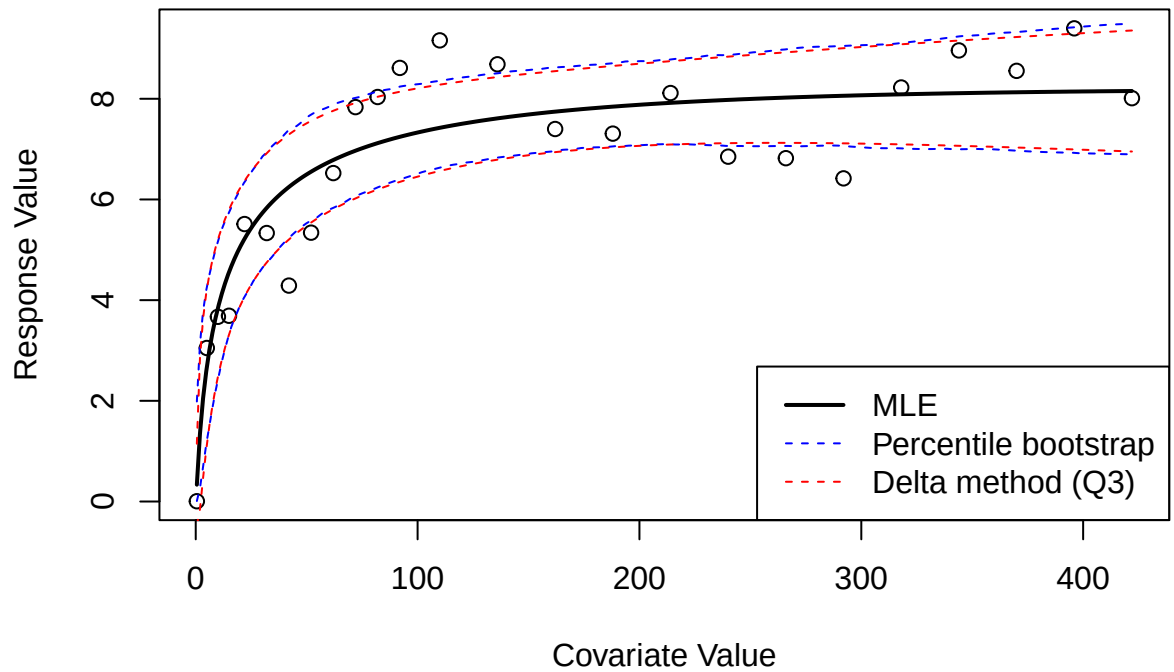
(b)

> Determine a 95% confidence band for the expectation function, also using percentile bootstrap methodology with maximum likelihood estimation.

Methodology. We construct a **pointwise** 95% confidence band for the expectation function $\mu(x;\, \alpha, \beta, \gamma) = \alpha x^{\beta x^{-\gamma}}$, meaning the nominal 95% coverage applies at each fixed x_0 separately, not simultaneously over the full curve. The figure below shows three curves: the MLE-centered fit, the percentile bootstrap band, and the delta-method band (described below), all evaluated on a grid of 300 equally spaced x-values spanning the observed covariate range.

For each retained bootstrap replicate $m = 1, \ldots, M_{\text{eff}}$, the re-estimated parameters $(\hat{\alpha}^*_m, \hat{\beta}^*_m, \hat{\gamma}^*_m)$ from part (a) are used to evaluate the fitted curve over a grid of 300 equally spaced x-values spanning the observed covariate range, yielding $\mu^*_m(x_0) = \hat{\alpha}^*_m\, x_0^{\hat{\beta}^*_m x_0^{-\hat{\gamma}^*_m}}$ at each grid point x_0. Following expression (10.15), the pointwise 95% percentile interval at x_0 is the empirical 2.5% and 97.5% quantiles of $\{\mu^*_m(x_0) : m = 1, \ldots, M_{\text{eff}}\}$, computed via interpolation since $(M_{\text{eff}} + 1) \cdot 0.025 = 24.725$ is not an integer. Sweeping this across all grid points produces the band shown in the figure.

For comparison, a delta-method band is also shown. It propagates uncertainty through the analytic gradient $\mathbf{d}(x_0) = \partial\mu/\partial(\alpha, \beta, \gamma)$ — the gradient of the expectation function with respect to the three mean parameters — via $\hat{V}(\hat{\mu}(x_0)) = \mathbf{d}(x_0)^T \hat{\Sigma}\, \mathbf{d}(x_0)$, where $\hat{\Sigma}$ is the (α, β, γ) block of the full 5×5 inverse observed Hessian from the Q3 joint estimation. Both bands are centered on the same MLE curve, so visible differences reflect methodology, not point estimation. In summary: the bootstrap band is asymmetric and wider in regions of sparse data, while the delta-method band is symmetric and narrower throughout due to its reliance on a linear approximation.



2.

Another possibility that we have at our disposal is to approach the problem through the use of Bayesian methods. Fit the model with extended Freundlich expectation function and autoregressive error structure using Bayesian analysis. To make this a manageable assignment, I will give some additional guidance. First, make use of the following prior distributions:

- For α, use a normal prior with mean 0.50 and variance 10, but truncate this prior at 0 to enforce the parameter space.
- For β, use a gamma prior with parameters 0.01 and 0.01 parameterized to give an expected value of 1 and a variance of 100.
- For γ, use an exponential prior with parameter 15. Note that this prior is quite informative and places probability 0.98 below a value of 0.25. This prior is chosen based on the rather extreme effect of the parameter γ on the response function.
- For θ, use a Laplace (or double exponential) prior with parameter 0.35, truncated below at -1 and above at 1. The value of 0.35 in a nontruncated Laplace distribution results in probability of 0.95 between -1 and 1, so the truncation is not severe, but we might want to preclude (eliminate) the possibility of a non-stationary error process for the w_i.
- For σ (not σ^2), use a uniform prior on the interval $(0, 20)$.

Use an overall Gibbs algorithm with several Metropolis pieces, one for the parameters α and β together, one for the parameter γ by itself, and the third for the parameters θ and σ^2 together. Note that, regardless of whether you express the likelihood and posterior in terms of σ^2 or σ, the uniform prior is assigned to σ.

As part of your answer, you should execute and report on the following:

(a)

Using the several short chains approach, determine an appropriate burn-in period based on an examination of the parameter α. You do not need to compute the Gelman-Rubin scale reduction factor, but you should present a trace plot with 3 chains started at different (as dispersed as you can reasonably manage) starting values. Also present a plot of autocorrelations for values of α. Note that in an actual problem of this type we might want to do these things for each of the parameters, but for this exam we will restrict attention to α.

Analytical justification

Target posterior. By Bayes' theorem the posterior is proportional to the product of the likelihood and the prior:

$$p(\alpha, \beta, \gamma, \theta, \sigma \mid \mathbf{y}) \;\propto\; L(\alpha, \beta, \gamma, \theta, \sigma^2 \mid \mathbf{y}) \cdot \pi(\alpha) \cdot \pi(\beta) \cdot \pi(\gamma) \cdot \pi(\theta) \cdot \pi(\sigma)$$

The likelihood with AR(1) errors $\varepsilon_i = w_i - \theta w_{i-1}$, $w_i = y_i - \mu(x_i; \alpha, \beta, \gamma)$, is

$$L \;=\; (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} \varepsilon_i^2\right)$$

This is the full conditional likelihood, treating $w_0 = 0$; the exact form would additionally include the marginal $w_1 \sim N(0, \sigma^2/(1-\theta^2))$.

Truncated normal for α. The untruncated $N(0.5, 10)$ density is $\phi\left(\frac{\alpha - 0.5}{\sqrt{10}}\right)/\sqrt{10}$. Truncating to $(0, \infty)$ renormalizes by the probability mass above zero:

$$\pi(\alpha) = \frac{\phi\left(\frac{\alpha-0.5}{\sqrt{10}}\right)/\sqrt{10}}{1 - \Phi\left(\frac{0-0.5}{\sqrt{10}}\right)} \cdot \mathbf{1}(\alpha > 0)$$

In the MH ratio this normalizing constant cancels when both current and proposed values satisfy $\alpha > 0$ — which is why returning $-\infty$ for $\alpha \le 0$ is sufficient and no explicit constant is needed in the code.

Truncated Laplace for θ. The Laplace$(0, 0.35)$ density is $\frac{1}{2(0.35)} \exp(-|\theta|/0.35)$. Truncating to $(-1, 1)$ gives

$$\pi(\theta) = \frac{\exp(-|\theta|/0.35)}{2(0.35)\left(1 - e^{-1/0.35}\right)} \cdot \mathbf{1}(|\theta| < 1)$$

Again the normalizing constant cancels in the MH ratio, so only $-|\theta|/0.35$ enters `log_prior`, with hard rejection for $|\theta| \ge 1$.

Log-posterior. For computational stability all evaluations are performed on the log scale. The unnormalized log-posterior is

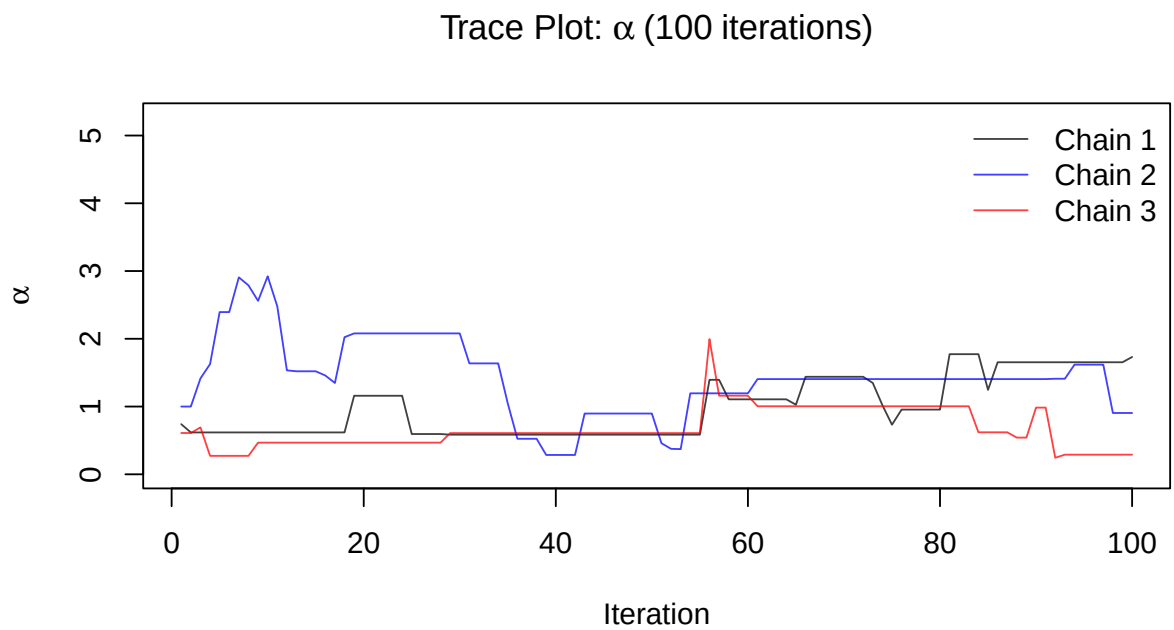
$$\log p(\boldsymbol{\psi} \mid \mathbf{y}) = \log L + \log \pi(\alpha) + \log \pi(\beta) + \log \pi(\gamma) + \log \pi(\theta) + \log \pi(\sigma)$$

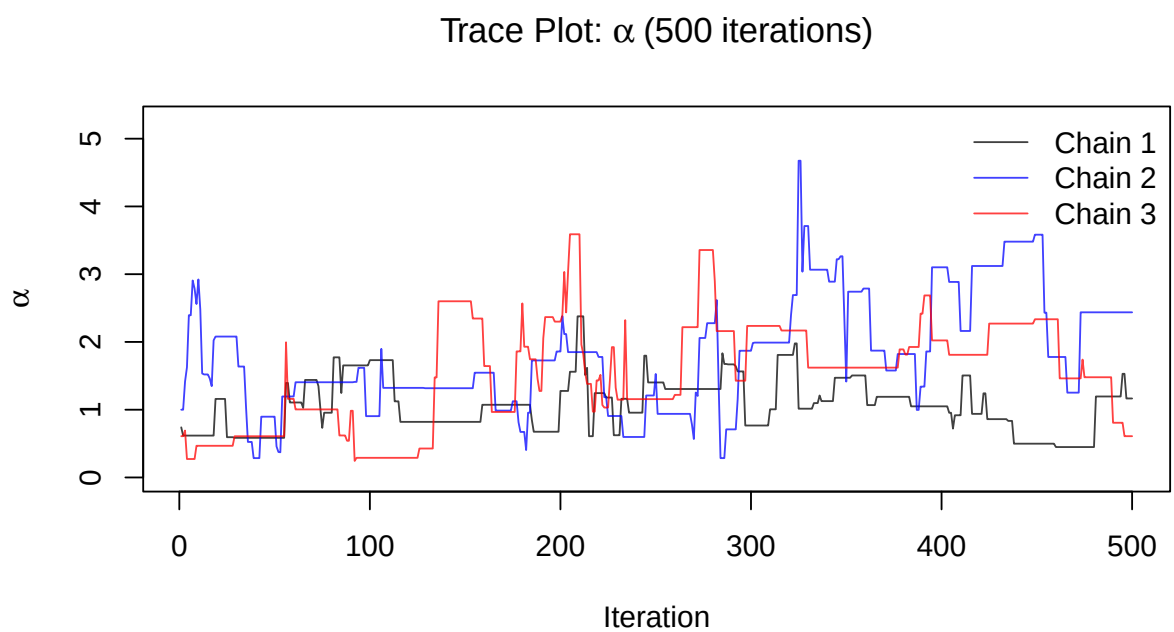
with the likelihood contributing
$-\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_i \varepsilon_i^2$.

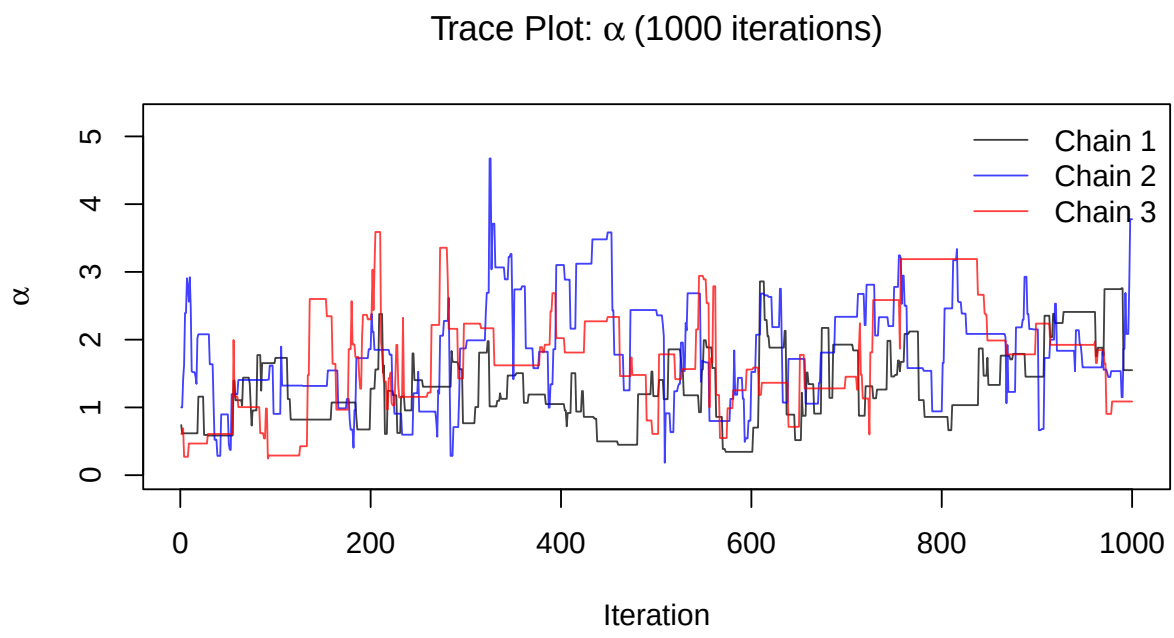
At each Metropolis step a proposal $\boldsymbol{\psi}^*$ is accepted with probability $\min\left(1,\, e^{\log p(\boldsymbol{\psi}^* \mid \mathbf{y}) - \log p(\boldsymbol{\psi}^{(t)} \mid \mathbf{y})}\right)$, evaluated in code as `log(runif(1)) < lp_proposed - lp_current`.

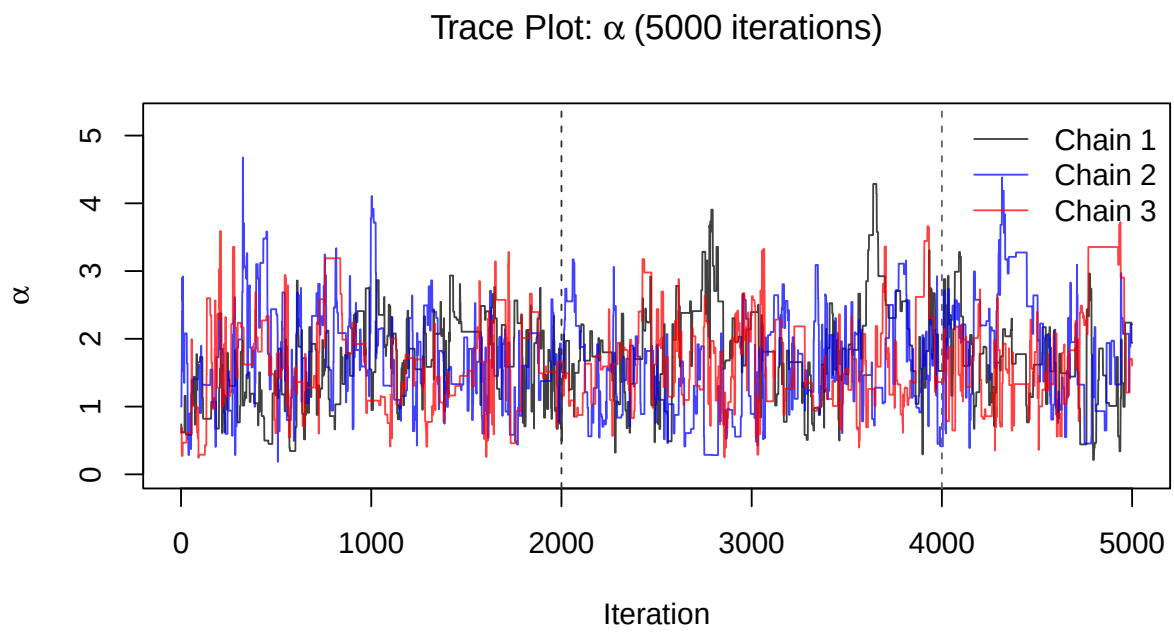
Why the uniform prior on σ, not σ^2. The prior $\pi(\sigma) \propto 1$ on $(0, 20)$ is not the same as $\pi(\sigma^2) \propto 1$; by the change-of-variables formula $\pi(\sigma^2) \propto 1/\sigma$, which is a more diffuse (Jeffreys-like) prior on the variance scale. The problem specifies the uniform on σ explicitly, so the posterior is sampled in σ and converted to σ^2 inside the likelihood evaluation — exactly what the code does.

Computation

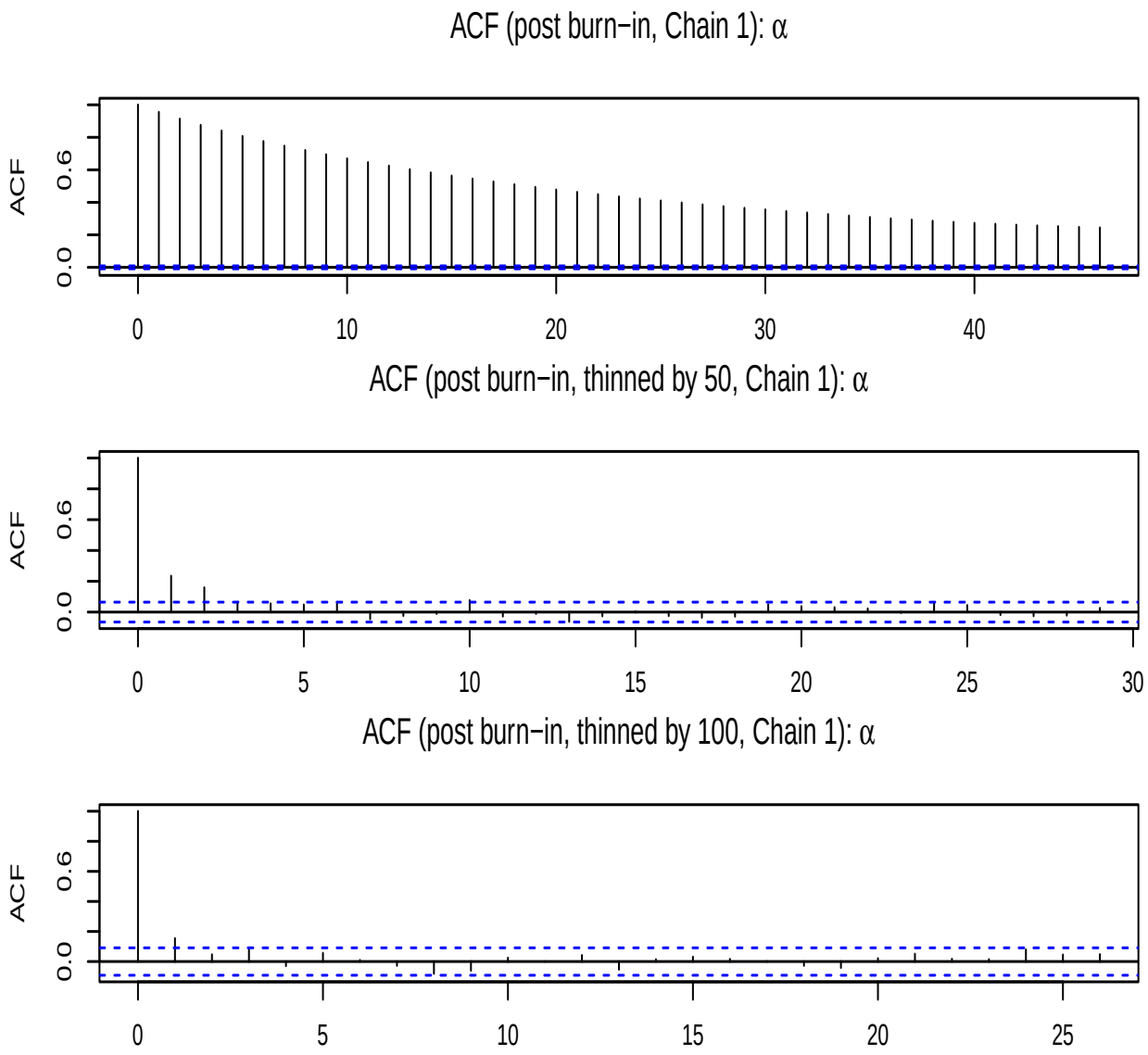








A burn-in of 4,000 iterations is selected. Three chains were initialized from dispersed starting values and run for several short trial lengths to justify this choice. The evidence proceeds as follows: trace plots at 100 and 500 iterations show clear between-chain separation, with chains traversing distinct regions of the parameter space. By 1,000 iterations the chains overlap substantially. By 2,000 iterations behavior is visually indistinguishable across chains — accounting for the compression inherent in plotting longer sequences. This 2,000-iteration stabilization point (dark grey line) is taken as the conservative convergence threshold; burn-in is set to double that value, 4,000 iterations (light grey line).



The raw post-burn-in ACF for α (top) reveals substantial autocorrelation that remains well above significance bounds even at lag 50, indicating slow mixing. Thinning by a factor of 100 (bottom) reduces autocorrelation to negligible levels beyond lag 1; thinning is used here diagnostically to characterize mixing, not as a required step for inference. The slow mixing is attributable to the high correlations among α, β, and γ (Table 2) that are not fully resolved by block-wise proposals. It is worth being explicit about the methodological implication: the several short chains approach was used here for its intended purpose — diagnosing burn-in by examining between-chain behavior across multiple dispersed initializations. However, the ACF evidence makes clear that short chains of the length adequate for burn-in assessment would yield very small effective sample sizes for inference. The posterior summaries in part (c) are therefore drawn from a single long chain of 54,000 iterations with the first 4,000 discarded, a deliberate departure from the short-chains framework motivated by the mixing behavior documented above. The effective sample size is considerably smaller than the nominal 4.6×10^4 retained draws, but the thinned ACF confirms that draws separated by 100 iterations are effectively independent, providing a usable posterior sample.

Table 2: Estimated correlation matrix of MLEs

	alpha	beta	gamma
alpha	1.000	-0.968	-0.685
beta	-0.968	1.000	0.843
gamma	-0.685	0.843	1.000

(b)

The proposal densities should all be random walks, and you will need to tune the Metropolis portions of your overall algorithm. Begin with a random walk variance of 0.50 for all parameters and tune the algorithm through trial and error to hopefully achieve acceptance rates of somewhere in the range of about 0.20 to 0.50. Present a table that shows variance specifications and acceptance rates. You do not need to show every step you used in your trial and error process, but show at least the beginning, the end, and one or maybe two places in between.

Table 3: Metropolis tuning: proposal SDs and acceptance rates.

						Acceptance Rate		
Step	σ_α	σ_β	σ_γ	σ_θ	σ_σ	Block 1	Block 2	Block 3
Initial	0.50	0.50	0.50	0.50	0.50	0.072	0.032	0.126
Attempt 2	0.80	0.80	0.05	0.50	0.50	0.038	0.255	0.120
Attempt 3	0.30	0.30	0.03	0.35	0.35	0.109	0.351	0.225
Final	0.10	0.10	0.03	0.25	0.25	0.286	0.348	0.329

Note:
Block 1: (α, β) jointly. Block 2: γ. Block 3: (θ, σ) jointly.

Table 3 reports the initial, two intermediate, and final tuning configurations. All tuning is conducted on the standard deviation scale (not variance); corresponding variances appear in the table footnote. Acceptance rates at each stage were averaged across all three chains over 2,000 iterations to avoid results being an artefact of any single run.

To start, Block 2 (γ) had an acceptance rate near 0.032 — far below target — because the informative Exp(15) prior concentrates γ tightly near zero, making large proposals almost always land in low-probability territory. The γ SD was reduced to 0.05 immediately. Block 1 (α, β) and Block 3 (θ, σ) were both below target, so their SDs were increased. This overcorrected Block 1 (0.038), so in Attempt 3 the SDs were pulled back to 0.3 and 0.35 respectively, with γ refined to 0.03. Block 1 remained below target (0.109), motivating the final reduction. T

The final proposal specifications achieve acceptance rates of 0.286, 0.348, and 0.329 for Blocks 1, 2, and 3 respectively, all within the target range of 0.20–0.50.

(c)

Collect $M = 50{,}000$ samples from the joint posterior after burn-in. Summarize the results using five number summaries, 95% credible intervals, and correlations in the joint posterior. Compare the point estimates and confidence intervals to what was obtained previously for maximum likelihood using a (probably incorrect) information matrix and using parametric bootstrap.

Posterior summaries and 95% credible intervals are reported in Tables 4–6; posterior correlations appear in Table 6. The discussion proceeds parameter by parameter, followed by an overall comparison.

α: The most striking discrepancy. The posterior median (1.692) is more than double the MLE (0.740), and the Bayesian credible interval (0.309, 3.327) is shifted substantially rightward relative to both the bootstrap (0.068, 2.511) and AN intervals. The AN interval is demonstrably invalid — it extends to -0.483, which violates the hard constraint $\alpha > 0$. This is the clearest illustration of why the information matrix was described as "probably incorrect" — the asymptotic normal approximation takes no account of the parameter boundary.

β: Posterior median (0.474) is roughly half the MLE (1.028). The Bayesian interval (0.181, 1.574) is notably narrower and lower than the bootstrap (0.309, 2.501). Due to strong posterior correlation ($\rho_{\alpha\beta} = -0.798$), combinations of higher α and lower β yield similar fitted values; the marginal MLE captures neither the shift nor the tradeoff.

γ: The informative Exp(15) prior is visibly pulling the posterior median (0.101) below the MLE (0.157), and the Bayesian interval (0.008, 0.202) is narrower than the bootstrap (0.055, 0.210). This is expected and by design — the prior was explicitly chosen to constrain γ based on subject-matter knowledge about its extreme effect on the response function.

θ: The posterior median (0.620) is modestly above the MLE (0.560), and the Bayesian interval (0.220, 0.957) sits entirely in positive territory. The bootstrap interval (-0.129, 0.649) includes negative values, suggesting the Laplace prior with its concentration near zero is providing meaningful regularization away from the lower tail.

σ^2: The posterior median (0.868) is notably higher than the MLE (0.625), and the Bayesian credible interval (0.481, 1.714) is considerably wider than both the bootstrap (0.263, 0.879) and AN intervals (0.279, 0.971). This makes sense — the Bayesian analysis propagates uncertainty from all other parameters into the marginal posterior for σ^2, whereas the MLE treats the other parameters as fixed at their point estimates.

Overall: Discrepancies between methods are largest where priors are informative or constraints are active (α, θ), and smallest where the likelihood dominates (γ). The AN intervals are least trustworthy throughout — most visibly for α, but the posterior correlation structure suggests the information matrix was poorly conditioned in ways that affect all five parameters.

Table 4: Five-number summary of posterior samples ($M = 50{,}000$).

Parameter	Min	Q1	Median	Q3	Max
α	0.0339	1.1537	1.6923	2.3007	4.5775
β	0.0012	0.3348	0.4743	0.6869	2.0691
γ	0.0000	0.0611	0.1007	0.1434	0.4249
θ	-0.1373	0.4854	0.6204	0.7551	0.9999
σ^2	0.2937	0.6982	0.8684	1.0897	4.5113

Table 5: Comparison of 95% intervals: Bayesian credible, percentile bootstrap, and asymptotic normal.

		Bayes		Bootstrap		Asym. Normal		
Parameter	Post. Med.	2.5%	97.5%	2.5%	97.5%	2.5%	97.5%	MLE
α	1.6923	0.3088	3.3273	0.0675	2.5107	-0.4834	1.9625	0.7395
β	0.4743	0.1811	1.5737	0.3085	2.5010	0.0275	2.0291	1.0283
γ	0.1007	0.0082	0.2024	0.0552	0.2099	0.0929	0.2219	0.1574
θ	0.6204	0.2196	0.9572	-0.1289	0.6488	0.2203	0.8993	0.5598
σ^2	0.8684	0.4808	1.7139	0.2633	0.8790	0.2785	0.9714	0.6250

Table 6: Posterior correlations among parameters ($M = 50{,}000$ samples).

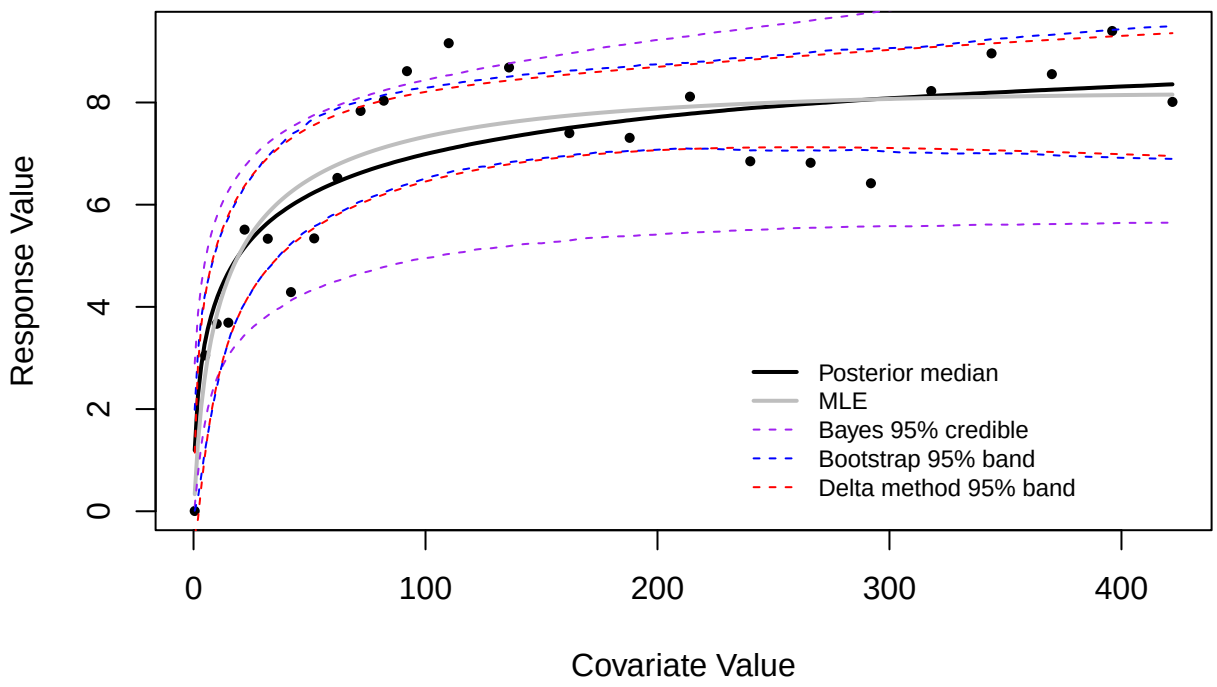
	α	β	γ	θ	σ^2
α	1.0000	-0.7977	-0.4599	0.0300	0.2474
β	-0.7977	1.0000	0.7507	-0.0344	-0.2265
γ	-0.4599	0.7507	1.0000	0.0551	-0.2142
θ	0.0300	-0.0344	0.0551	1.0000	0.0143
σ^2	0.2474	-0.2265	-0.2142	0.0143	1.0000

(d)

A scatterplot of the data with the median expectation function and a pointwise 95% credible band. Why might we prefer using the posterior medians here rather than the posterior means? Compare this Bayesian credible band to those constructed using likelihood theory, both asymptotic normality and as constructed by parametric bootstrap.

Posterior medians vs. means: Posterior medians are preferred here due to skewness and the hard boundary constraint on α. The posterior distributions of α and β are right-skewed — visible in the five-number summaries where Q3 − Median > Median − Q1 for both parameters — and α has a hard lower boundary at zero imposed by the truncated normal prior. The posterior mean is sensitive to skewness and to occasional large draws in the upper tail, making it a less stable summary of central tendency. The median is more robust to this skewness. And importantly, the plug-in curve $\mu(x; \tilde{\alpha}, \tilde{\beta}, \tilde{\gamma})$ evaluated at the posterior medians $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma})$ is not the same as the pointwise posterior median of $\mu(x)$ — but it provides a single coherent curve rather than a parameter-by-parameter summary that may not correspond to any consistent parameter vector. For a nonlinear function like the extended Freundlich, the pointwise posterior median of $\mu(x)$ computed directly from the Monte Carlo draws — as done here — is generally preferred over plugging in marginal means for two related reasons: the marginal means may not correspond to any coherent parameter vector given the strong posterior correlations documented in Table 6, and the nonlinearity of μ means that $E[\mu(x; \boldsymbol{\psi})] \neq \mu(x; E[\boldsymbol{\psi}])$ in general. Both concerns favor working directly with the Monte Carlo distribution of the curve rather than with summaries of the marginal parameter distributions.

Comparing bands: The most immediately striking feature of the plot is the separation between the two central curves — the posterior median curve sits visibly above the MLE throughout the middle of the covariate range, a direct consequence of the upward shift in the marginal posterior of α relative to its MLE documented in part (c). Regarding the bands themselves, the Bayesian credible band is the widest of the three at virtually every covariate value, and notably so at high covariate values where data are sparse — reflecting joint uncertainty across all five parameters via the posterior distribution. The bootstrap band is intermediate in width and shares the Bayesian band's asymmetry, since both respect the nonlinearity of the expectation function rather than relying on a linear approximation. In summary: the Bayesian band is widest and asymmetric, accounting for full parameter uncertainty; the bootstrap band is intermediate and asymmetric, respecting the nonlinearity of μ without a linear approximation; the delta-method band is narrowest and symmetric, centered on the same MLE curve as the bootstrap band but underestimating uncertainty, most visibly at large covariate values where data are sparse.

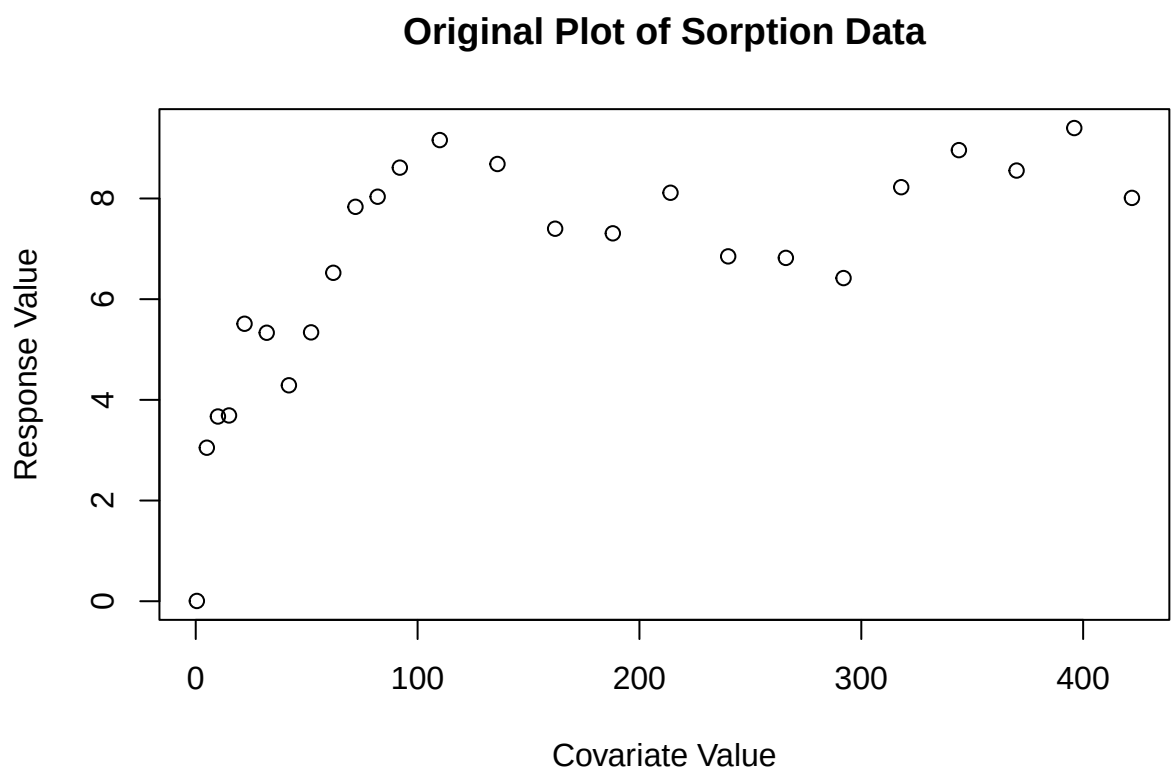

Response Value
0
2
4
6
8
0
100
200
300
400
Covariate Value
Posterior median
MLE
Bayes 95% credible
Bootstrap 95% band
Delta method 95% band

Misc Note:

Throughout Parts I and II, the expectation function remains fixed — a structural limitation distinct from, and not resolved by, the inferential challenges addressed at each stage. Increasing the number of parameters to include σ^2 and θ does not make the extended Freundlich more flexible in shape. The original sorption data (plotted below) exhibit apparent non-monotone behavior around covariate values 100 and 300 that no reparameterization of the extended Freundlich — nor the Fitter-Sutton equation ($m = Ac^B - D$), the next-best alternative noted in Sibbesen (1981) — can accommodate. Barrow (2023) makes a related point: empirical formulations of this family are approximations whose adequacy depends heavily on the concentration range and resolution of the data.

This observation may be grasping at straws. With $n = 23$ and no replicates, there is no way to formally distinguish genuine non-monotone behavior from noise — the apparent oscillations could plausibly be sampling variation. But the absence of replication is itself the point. When replicates exist, lack-of-fit tests make model adequacy a routine diagnostic. Here they do not, and a potentially important feature of the data passes without formal assessment. Whether or not the oscillations are real, the inability to test for them is a genuine limitation that the analysis does not acknowledge.

Implications for simulation assessment. A less obvious consequence follows. Both the parametric bootstrap in Part I and the Gibbs sampler in Part II use the fitted Freundlich as the data-generating process. If the oscillatory structure reflects genuine features of the sorption mechanism — rather than AR(1) dependence around a smooth mean — then both procedures generate data from a misspecified model, and the coverage properties of all reported intervals are only as reliable as that assumption. A more stringent evaluation would generate data from a flexible alternative and assess whether the Freundlich-based methods can detect the misspecification; that such an assessment is not possible within the current parametric framework is itself worth stating explicitly. Neither Part I nor Part II addresses the prior question on which both depend: whether the Freundlich family is adequate for these data to begin with.



Appendix - Gibbs Sampler

```r
# Log-likelihood under AR(1) error structure
log_lik <- function(alpha, beta, gamma, theta, sigma2, x, y, n) {
  mu <- alpha * x^(beta * x^(-gamma))
  w <- y - mu
  eps <- w - theta * c(0, w[-n])
  -(n/2)*log(2*pi) - (n/2)*log(sigma2) - sum(eps^2)/(2*sigma2)
}

# Log prior - unnormalized, truncation constants cancel in MH ratio
log_prior <- function(alpha, beta, gamma, theta, sigma) {
  if (alpha <= 0 || beta <= 0 || gamma <= 0 ||
      sigma <= 0 || sigma > 20 || abs(theta) >= 1) return(-Inf)
  lpa <- dnorm(alpha, 0.5, sqrt(10), log = TRUE) # N(0.5, 10) truncated at 0
  lpb <- dgamma(beta, shape = 0.01, rate = 0.01, log = TRUE) # Gamma(0.01, 0.01)
  lpg <- dexp(gamma, rate = 15, log = TRUE) # Exp(15)
  lpt <- -abs(theta) / 0.35 # Laplace(0, 0.35) truncated to (-1,1)
  lpa + lpb + lpg + lpt # Uniform(0,20) on sigma contributes constant
}

# Log posterior
log_post <- function(state, x, y, n) {
  alpha <- state[1]; beta <- state[2];
  gamma <- state[3]
  theta <- state[4]; sigma <- state[5]
  lp <- log_prior(alpha, beta, gamma, theta, sigma)
  if (!is.finite(lp)) return(-Inf)
  log_lik(alpha, beta, gamma, theta, sigma^2, x, y, n) + lp
}

# Single Gibbs sweep with three MH blocks; tracks acceptance counts
gibbs_step_tracked <- function(state, prop_sd, x, y, n, acc) {
  lp_c <- log_post(state, x, y, n)

  # Block 1: (alpha, beta) jointly
  prop <- state
  prop[1:2] <- state[1:2] + rnorm(2, 0, prop_sd[1:2])
  lp_p <- log_post(prop, x, y, n)
  if (log(runif(1)) < lp_p - lp_c) { state <- prop; lp_c <- lp_p; acc[1] <- acc[1] + 1 }

  # Block 2: gamma alone
  prop <- state
  prop[3] <- state[3] + rnorm(1, 0, prop_sd[3])
  lp_p <- log_post(prop, x, y, n)
  if (log(runif(1)) < lp_p - lp_c) { state <- prop; lp_c <- lp_p; acc[2] <- acc[2] + 1 }

  # Block 3: (theta, sigma) jointly
  prop <- state
  prop[4:5] <- state[4:5] + rnorm(2, 0, prop_sd[4:5])
  lp_p <- log_post(prop, x, y, n)
  if (log(runif(1)) < lp_p - lp_c) { state <- prop; acc[3] <- acc[3] + 1 }
```

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  list(state = state, acc = acc)
}

# Chain runner
run_chain_tracked <- function(start, n_iter, prop_sd, x, y, n) {
  mat <- matrix(NA, n_iter, 5,
                dimnames = list(NULL, c("alpha","beta","gamma","theta","sigma")))
  acc <- c(0, 0, 0)
  state <- start
  for (i in seq_len(n_iter)) {
    out <- gibbs_step_tracked(state, prop_sd, x, y, n, acc)
    state <- out$state
    acc <- out$acc
    mat[i,] <- state
  }
  list(chain = mat, accept_rates = acc / n_iter)
}

# Example: run chain from MLE starting values, 54000 iterations, discard first 4000
prop_sd <- c(alpha = 0.10, beta = 0.10, gamma = 0.03, theta = 0.25, sigma = 0.25)
burn <- 4000
n_iter_c <- burn + 50000

mle_sigma <- sqrt(mle["sigma2"])
start1 <- c(mle["alpha"], mle["beta"], mle["gamma"], mle["theta"], mle_sigma)

set.seed(42)
out <- run_chain_tracked(start1, n_iter_c, prop_sd, x, y, n)
post <- as.data.frame(out$chain[(burn + 1):n_iter_c, ])
post$sigma2 <- post$sigma^2
post$sigma <- NULL
```